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CASINO: A New Monte Carlo Code in C Language for Electron Beam Interactions—Part III : Stopping Power at Low Energies

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Summary: This paper is a description of the stopping power routine utilized in the *CASINO* program that is based on the experimental measurement of the energy loss function (ELF). In addition, we present an ANSI C standard program that can be used to generate the data needed for the stopping power routine. Both optical and energy loss spectrum (ELS) measurements of the ELF can be used as input to compute the stopping power. For ELS, only the single scattering spectrum is needed. Hence, measurement of the stopping power for a given element or compound of interest can easily be performed and used in the *CASINO* program. The resulting effect of using these stopping powers in Monte Carlo simulations is generally to increase the backscattering coefficient. Except for carbon, the change of stopping power for pure elements so far compiled is relatively small. In some compounds (i.e., Al_2O_3 and ZnSe), the discrepancy with the Joy and Luo (1989) expression is significant.

Key words: experimental stopping power, energy loss function, Monte Carlo simulation, low-energy scanning electron microscope, backscattered electrons

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Introduction

In conventional Monte Carlo programs using a continuous slowing down approximation, an accurate knowledge of the stopping power is critical since it determines the calculated electron range, both lateral and in depth, the energy transfer per mass thickness, and thus the depth distribution of inner shell ionization $\varphi(\rho z)$. At high energy ($E \geq 10$ keV), the Bethe (1930) equation is known to provide an accurate and simple analytical expression for the stopping power. This expression was derived for the case where all the available excitations are included (Joy and Luo 1989):

$$\frac{dE}{dS} = -\frac{7.85 \times 10^4 \cdot \rho}{E} \frac{CZ}{A} \ln\left(\frac{1.166E}{J}\right) \quad (keV / cm) \tag{1}$$

where C, J, and A, are, respectively, the weight fraction, the mean excitation energy, and the atomic weight (g/mole) of the element. Z is the number of effective electrons participating in the excitation up to the energy transfer E, which corresponds at high energy to the atomic number of the element. However, at low energy, some inelastic events (e.g., inner shell ionization) are no longer possible and, as a consequence, the value of the mean excitation energy (J) and the number of electrons contributing to energy loss up to an energy E (Z) must decrease accordingly.

At low energy, the stopping power can be computed from the imaginary part of the dielectric function obtained from either electron energy loss spectrum (ELS) measurement or optical measurement of the refractive index (both real and imaginary parts). However, no analytic expression is presently available to calculate the stopping power from these measurements. In this paper, we present a simple C routine that uses a tabulated form of stopping power based on dielectric measurements. This routine is presently available in the CASINO program (Hovington *et al.* 1997). In addition, we supply and describe a program that can be used to compute the stopping power from a single scattering ELS or optical measurement. A stand-alone program in C language will be briefly discussed. All source codes presented here are available on the web site “http://www.gme.usherb.ca/casino.”

Determination of the Stopping Power from a Single Scattering Spectrum

The most powerful technique for determining the stopping power at low energy is based on the use of an energy loss spectrometer (Joy *et al.* 1996). The energy loss process can be viewed as the dielectric response of the specimen to the passage of a charged particle. Hence, the ELS contains information about the dielectric constant or permitivity ($\varepsilon=\varepsilon_1+i\varepsilon_2$) of the material (Daniels *et al.* 1970). For small scattering angles, the single-scattering spectrum intensity, S(E), is related to the energy loss function (ELF) ($\mathrm{Im}[-1/\varepsilon(E)] \equiv \varepsilon_2/(\varepsilon_1^2+\varepsilon_2^2)$) of the specimen by (Egerton 1986)

$$S(E) = \frac{I_0 t}{\pi a_o m_o v^2} \mathrm{Im}[-1/\varepsilon(E)] \ln\left[1+(\beta/\theta_e)^2\right] \tag{2}$$

where I_o = intensity in the zero-loss peak, t = specimen thickness, a_o = bohr radius (0.0529 nm), m_o = electron mass (9.11×10^{-31} kg), v = incident-electron velocity, β = collection semi-angle, and θ_E = characteristic scattering angle for an energy loss ($= E/\gamma m_o v^2$).

The real part of the dielectric constant ($\mathrm{Re}[1/\varepsilon(E)]$) is extracted from the imaginary part with a Kramers-Kronig analysis (Egerton 1986). Unless the value of t is accurately known, $\mathrm{Im}[-1/\varepsilon(E)]$ can be given an absolute scale by the use of a Kramers-Kronig sum rule:

$$1 - \mathrm{Re}\left[\frac{1}{\varepsilon(0)}\right] = \frac{2}{\pi}\int_0^\infty \mathrm{Im}[-1/\varepsilon(E)]\frac{dE}{E} \tag{3}$$

Here, $\mathrm{Re}[1/\varepsilon(0)] = \lim_{E\to 0}\left[\frac{\varepsilon_1}{\varepsilon_1^2+\varepsilon_2^2}\right]$. In a metal, both ε_1 and ε_2 become large in the limit of low-energy loss and $\mathrm{Re}[1/\varepsilon(0)]$ tends to zero. In an insulator, ε_2 becomes very small at low energy and $\mathrm{Re}[1/\varepsilon(0)] \approx 1/\varepsilon_1(0) \approx 1/n_1^2$, where n_1 is the refractive index for visible light (Egerton 1986).

The Bethe f-sum rule (Bethe 1930) gives rise to the concept of an effective number of electrons [$Z_{eff}(E)$] contributing to the energy loss up to an energy E:

$$Z_{eff}(E) = 7.6598 \times 10^{-4} \rho \sum_i^n \left(\frac{C_i}{A_i}\right) \int_o^E \mathrm{Im}[-1/\varepsilon(E')dE'] \tag{4}$$

where C_i and A_i are the weight concentration and atomic weight of the element i and ρ is the density of the analyzed region. At high energy, when all the inelastic processes are summed, Z_{eff} should asymptotically reach Z, the total number of electrons in the analyzed region. This relation can be used as an additional normalization when ε_1 and/or ε_2 are not accurately known. Hence, the only necessary input for the program to compute the ELF is the single-scattering spectrum (without the zero loss intensity), the atomic number (Z), the atomic weight (A), and the density (ρ) at the analyzed region. The EL/P program of GATAN (1993) can be used to obtain the single-scattering spectrum.

For condensed materials, the mean effective excitation energy for an energy E [$J_{eff}(E)$] can be expressed in terms of the ELF of the medium (ICRU1984):

$$\ln(J_{eff}(E)) = \frac{\int_o^E \ln(E')\,\mathrm{Im}[-1/\varepsilon(E')]E'dE'}{\int_o^E \mathrm{Im}[-1/\varepsilon(E')]E'dE'} \text{(eV)} \tag{5}$$

Replacing the Z and J values of the conventional Bethe (1930) expression with the Z_{eff} and J_{eff}, we found:

$$-\frac{dE}{dS} = -\frac{785.}{E}\left[\frac{\rho C}{A}\right] Z_{eff}(E) \ln\left(\frac{1.166E}{J_{eff}(E)}\right) \quad (eV/A) \tag{6}$$

We present in Figure 1 the $Z_{eff}(E)$, $J_{eff}(E)$ for Al computed with an ELF obtained from ELS measurement. Abrupt changes in the slopes of both J_{eff} and Z_{eff} are clearly visible at the energies corresponding to both plasmon and core-loss excitation energies. The saturation value of J_{eff} is very close to the predicted value of 160 eV. Details and the physical basis of this procedure are given elsewhere.

The ELS was performed at 100 kV using a Philips EM400/FEG transmission electron microscope equipped with a GATAN 666 parallel-detection electron energy-loss spectrometer (PEELS). During microscopic examination, the sample (electropolished 1100 aluminum) was kept at –130°C to minimize specimen contamination under the electron beam. In addition, the field emission tip was operated cold during the time of acquisition (approximately 5 min for a complete set of spectra) to improve the energy res-

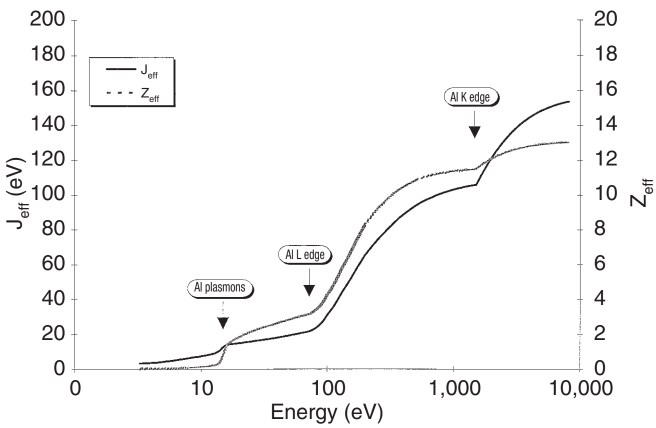

FIG. 1 Z_{eff} and J_{eff} for an 1100 Al sample. Eo=100 keV, β=5 mrad.

olution (FWHM = 0.6 eV for the zero-loss peak). Three spectra, with overlapping energy ranges, were acquired with an energy dispersion of 0.3 eV/channel. Those spectra were then spliced together with the GATAN EL/P software to provide to a single spectrum containing more than 1500 channels.

At high energy, the signal-to-noise ratio of the electron ELS is very weak and it is often better to use x-ray mass absorption coefficient (MAC, μ/ρ) to compute the dielectric imaginary loss function:

$$\left[\frac{\mu}{\rho}\right] = \frac{2E}{\hbar\rho c}\left[\frac{1}{2}\left(\sqrt{\varepsilon_1^2+\varepsilon_2^2}-\varepsilon_2\right)\right]^{1/2} \tag{7}$$

where $\hbar = h/2\pi$ (where h is the Planck constant), and c is the speed of light. At sufficiently high energy ($E > 150$ eV), ε_1 can be taken as unity. MACs are usually used for energy > 250 eV. For low-energy x-rays, MACs of Henke and Ebisu (1974) are used, subject to the modification proposed by Pouchou and Pichoir (1991). At higher energy ($E \geq 1$ keV) the formulation of Heinrich (1987) is preferred.

In Figure 2, we present the stopping power for Al, computed from two experimental ELS spectra (β=5 and 100 mrad), optical measurements, a calculation of Fernandez-Vea *et al.* (1993) and the measurements of Al-Ahmad and Watt (1983). The difference between the stopping powers acquired with different collection angles has been attributed to the relationship between mean energy loss and the characteristic scattering angle of the incident electron.

Determination of the Stopping Power from Optical Measurements

The ELF (-Im[1/ε]) can also be obtained from light reflection measurements at well-prepared surfaces over a wide region of wavelengths. The optical dielectric constant, ε, is related to the refractive index n and the extinction coefficient, k by

$$\varepsilon(E) = (n+ik)^2 = \varepsilon_1 + i\varepsilon_2 \tag{8}$$

so that we have $\varepsilon_1 = n^2 - k^2$ and $\varepsilon_2 = 2nk$. Optical data (n and k) have been compiled for a number of materials (Palik 1985, 1990). Experimentally, no significant differences were found in the ELF obtained from ELS and optical measurements even though electron excitations correspond to a longitudinal electromagnetic wave displacement compared with transverse wave displacement for light (Daniels *et al.* 1970).

At high energy, the optical method may run into difficulties because of the strong dependency on the surface conditions and the need to use high-energy synchrotron radiation. As a result, the tabulated optical data do not extend to sufficiently high energies to include all inner shell ionizations. Again, we used the mass absorption coefficient to compute the ELF at these high energies.

We present in Figure 3 the ELF of Au computed from optical measurements (Palik 1990). Also presented is the ELF computed with the ELS of Wehenkel (1975) combined with x-ray MAC for energies > 200 eV. Clearly, good agreement is found between the ELF computed from the x-ray MAC and the optical measurements.

Description of the STOPPING.C Program

The STOPPING program is used to compute electron stopping power from ELS or optical measurements. The flow chart of the program is presented in Figure 4.

As inputs, the program requires the type of data (ELS or optical), the number of elements in the analyzed region, and weight fraction of each element.

For outputs, two files are created: an ASCII file which contains, in order, the energy, $Z_{eff}(E)$, $J_{eff}(E)$, the resulting stop-

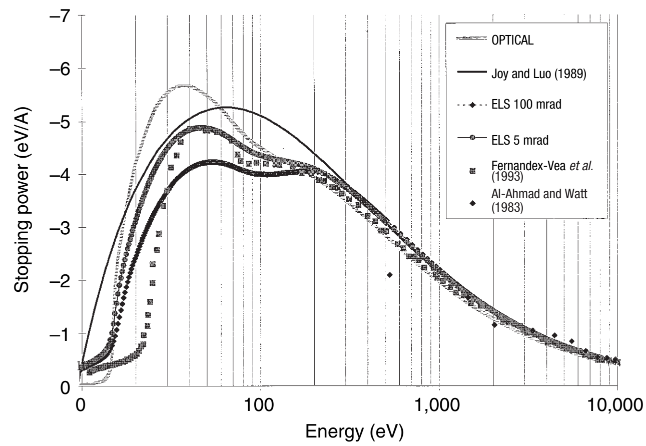

FIG. 2 Stopping power of Al computed from optical data (Palik 1990), ELS measurement (β=5 and 100 mrad), calculated from Fernandez-Vea *et al.* (1993), and measure of Al-Ahmad and Watt (1983).

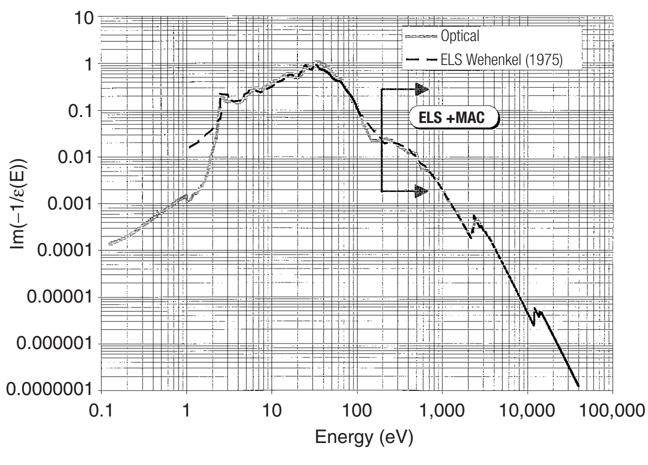

FIG. 3 Energy loss function of Au computed from optical data (Palik 1990) and the energy loss spectrum (ELS) of Wehenkel (1975). For ELS computation and for an energy > 200 eV the x-ray MAC was used.

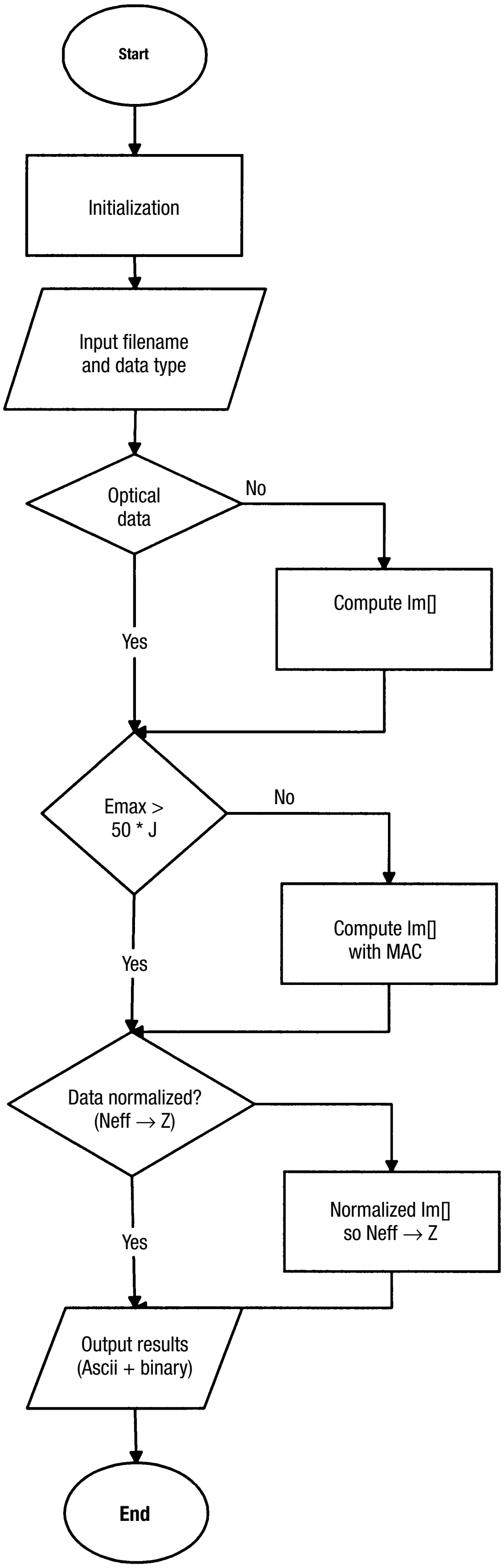

FIG. 4 Flowchart of the STOPING.C program.

ping power (in eV/Å), and the stopping power computed from Joy and Luo (1989). This file can be easily imported into any database program for further examination.

The binary output file is the data file requested by the *dEdS_Fich* routine of the *CASINO* program. This file contains the stopping power in keV/[nm-(g/cm^3)] from 5 eV to 30 keV with a logarithmic energy step. Two hundred (200) points are used to model the experimental stopping power (from 5 eV to 30 keV). The *spline* routine of Press *et al.* (1992) is used for data modeling. For energies > 50*J, Eq. (1) is used.

Both the ASCII and the binary files have formated filenames. The ASCII filename is always EXPSPWR.ZZ. The extension (ZZ) is the atomic number for pure elements and the mean atomic number for compounds. For binary files, the same convention as with the total and partial elastic cross section is used (see Drouin *et al.* 1997). For optical stopping power, the filename is of the form oXY.SPT, where XY correspond to the chemical symbol. For elements with only one character, X and Y are the same character. For the stopping power computed with an ELS, the o is replaced by an e (i.e., eXY.SPT). In the case of compounds, the filename must be typed by the user.

Description of CASINO Stopping Power Routine

We present in Figure 5 the flowchart of the *dEdS_Fich* subroutine. The parameters of this routine are the region identification number of the collision and the energy at which the stopping power must be computed. The procedure will return the stopping power in keV/nm. To decrease the access time of this routine, all the necessary information, namely, the stopping power of each element and region of the sample, are entered into memory at the first call (first trajectory loop). Following the first calls, the only process involved in this procedure is to compute the vector position corresponding to the energies just lower and just higher than the energy requested by the main loop. With this minimum and maximum value, the program interpolates the stopping power and multiplies the results by the density of the region.

For compounds, a similar subroutine (*dEdS_Fich_Comp*) is used. In this routine, the filename of the compound is entered before compiling the program. The execution scheme is similar to the *dEdS_Fich* routine (cf. Fig. 5.)

Description of the Standalone Program ("SPFICH.C")

A stand-alone routine (SPFICH) that computes the stopping power [in keV/(nm-(g/cm^3))] from the binary data files output by the program "STOPPING" can be found at the following web site: "http://casino.gme.usherb.ca/casino." The input parameters of the routine are the atomic number Z of the element and the energy E (keV) at which the stopping power must be computed. The resulting stopping power must be multiplied by the density of the analyzed region. This program has the same features (SYSTEM and DATALOCATION) as the "Sec_Total_MottFich" routine (Drouin *et al.* 1997).

Speed Test

When we compare this routine with the Eq. (1) expression we find a small increase of approximately 10% in computing time. This increase is mainly due to the use of a logarithmic energy scale that decreases the amount of memory usage but increases the computing time. Having the STOPPING procedure and the CASINO source files, it is then straightforward to modify this routine so that a linear energy scale is used. With these modifications, this routine should be even faster than a subroutine based on Eq. (1).

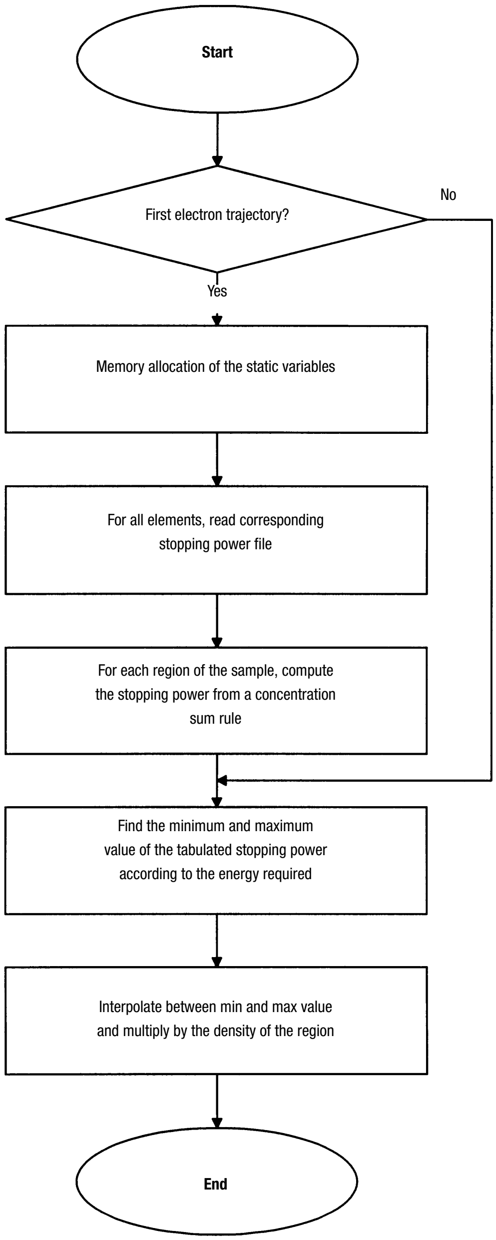

FIG. 5 Flowchart of the CASINO *dEdS_Fich* subroutine.

Examples of Application

We present in Figure 6 the backscattering coefficients of C, Al, Pt, and Al_2O_3 computed with the *CASINO* program using the conventional Joy and Luo (1989) expression and the experimentally obtained stopping power. For all the simulations, an interpolated Mott elastic cross section was used (Drouin *et al.* 1997). We also present the resulting backscattering coefficient when the Rutherford cross section is used with the stopping power of Joy and Luo (1989). A larger difference is observed for carbon. However, negligible effects are found for Al and Pt. For the Al_2O_3 compound, as expected (cf. Fig. 9), a large discrepancy is also observed. No experimental value was found to confirm these results.

We present in Figure 7 the backscattering profile from an 88° Si edge analyzed at 0.7 keV. The edge step is 1 µm in height. This sample was used for the National Institute of Standards and Technology Monte Carlo metrology round robin (Lowney *et al.* 1995). In Figure 7, the backscattering profile obtained with the Joy and Luo (1989) expression and the experimental stopping power are both plotted. The differences between the two profiles are, in general, negligible.

We compare in Figure 8 the stopping power of Si computed with the Joy and Luo (1989) expression and the one computed from optical measurements. The difference between the two stopping powers is significant only for energies < 90 eV. The depth at which an electron would reach this energy is, generally, great enough so that the escape probability of such an electron is small. However, if the incident beam is near an edge wall, low-energy electrons can escape the sample. As a consequence, the most significant differences between the two profiles are found near the edge (cf. Fig. 7).

Nevertheless, in some other cases, the differences between experimentally determined stopping power and the Joy and Luo (1989) expression are significant. We present in Figure 9 the stopping power of alumina (Al_2O_3) computed from optical measurement and the Joy and Luo (1989) expression. The relative differences here are greater than in the case of pure Al (cf. Fig. 2). The discrepancy has been attributed to the large difference in atomic bonding between metallic aluminum and alumina.

For other compounds such as silica (SiO_2), the difference between the Joy and Luo (1989) expression and the experimental stopping power is less pronounced (Joy *et al.* 1995), as shown in Figure 10. The number of compounds and corresponding pure elements is still too low to draw a conclusion as to any systematic deviation between the Joy and Luo (1989) expression and the experimental data. A systematic program of study is still required to generate data to cover all the materials of interest (Joy *et al.* 1996).

Conclusions

Electron stopping powers calculated from experimental measurements of energy loss function are now becoming available. These stopping powers take into account all

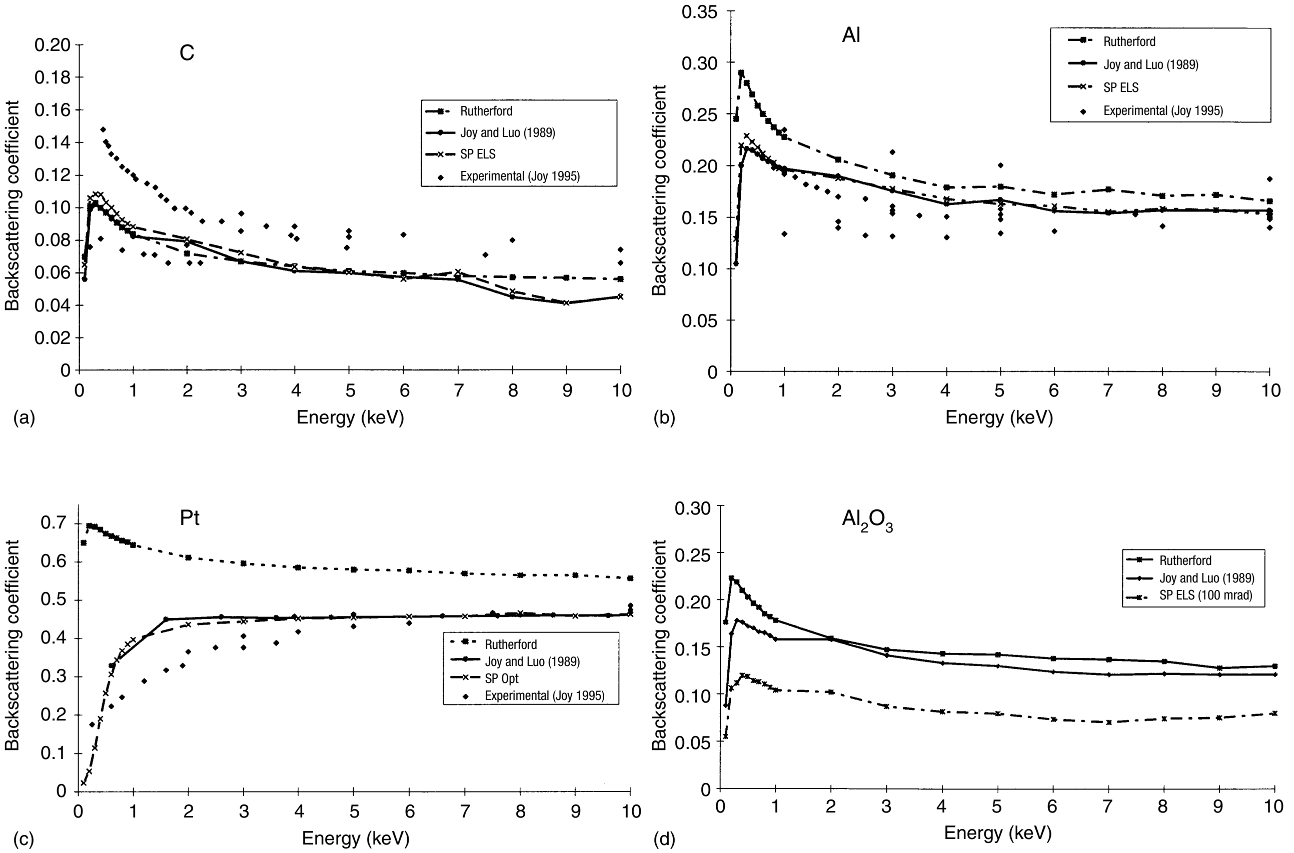

FIG. 6 Backscattering coefficient as a function of energy of C, Al, Pt, and Al_2O_3 computed with the Joy and Luo (1989) expression and simulated with CASINO using experimentally determined stopping power (optical or energy loss spectrum). Backscattering coefficient using the Rutherford elastic cross section and the Joy and Luo (1989) expression is also presented. The experimental values can all be found in the Joy (1995) data base. 250,000 electrons were used for each simulation point.

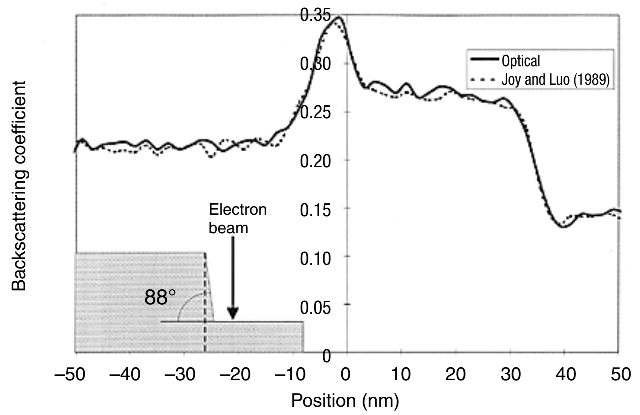

FIG. 7 Backscattering profile from an 88° Si edge (1 mm height) analyzed at 0.7 keV.

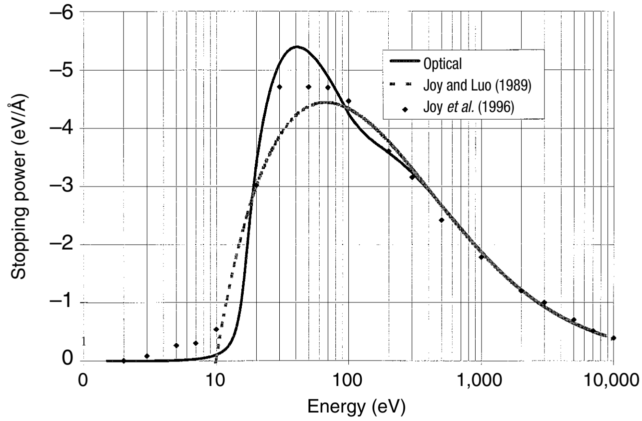

FIG. 8 Stopping power of Si computed from the Joy and Luo (1989) expression and determined experimentally from optical measurement (Palik 1985) and energy loss spectrum measurement (Joy *et al.* 1996).

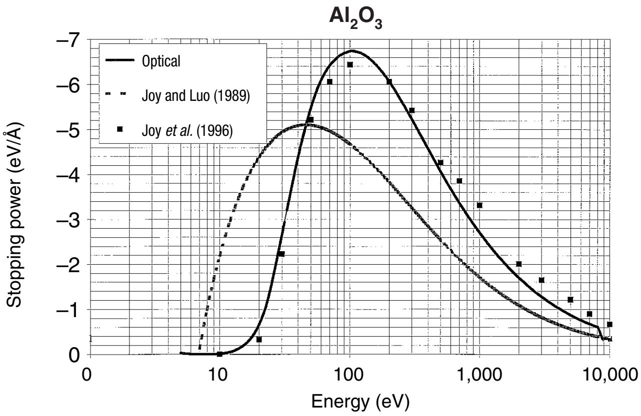

FIG. 9 Stopping power of Al_2O_3 computed from Joy and Luo (1989) expression and determined experimentally from optical measurement (Palik 1990) and energy loss spectrum measurement (Joy *et al*. 1996).

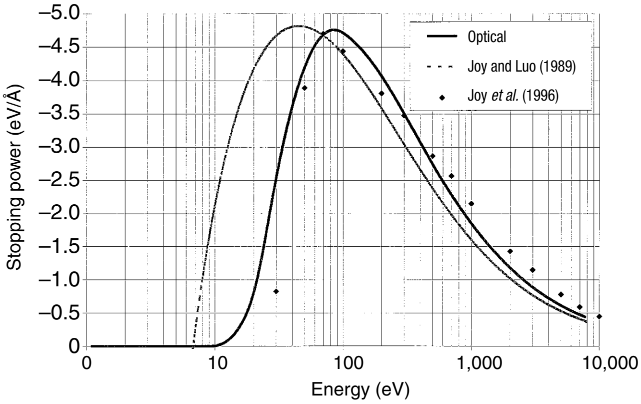

FIG. 10 Stopping power of SiO_2 computed from Joy and Luo (1989) expression and determined experimentally from optical measurement (Palik 1990) and energy loss spectrum measurement (Joy *et al*. 1996).

mechanisms of electron energy loss in the element or compound of interest. They can cover the full range of energies used in the field-emission gun scanning electron microscope (0.1–30 keV). Combining these stopping powers with tabulated elastic Mott cross sections in a Monte Carlo program permits a simulation engine fully adapted to energies down to 50 eV. The resulting effects of using these stopping powers are generally increased backscattering coefficients compared with conventional calculation. Except for carbon, the effect on the pure elements so far compiled is relatively small. In some compounds (i.e., Al_2O_3), the discrepancy with the Joy and Luo (1989) expression is significant and important effects will occur on all the simulated output [$\varphi(\rho z)$, range, energy distribution of backscattering electrons, etc]. However, in some other compounds (i.e., SiO_2), the experimental stopping powers are comparable to the Joy and Luo (1989) expression. Studies of additional compounds and pure elements are currently under way and aim to cover a greater portion of the Periodic Table.

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